



Republic of the Philippines
CEBU TECHNOLOGICAL UNIVERSITY
MAIN CAMPUS
M. J. Cuenco Avenue Cor. R. Palma Street, Cebu City, Philippines
Website: <http://www.ctu.edu.ph> E-mail: ccictdean@ctu.edu.ph
Phone: +6332 402 4060 loc. 1104



COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

PC 317
Systems Analysis and Design

Worksheet No. 4: Technical Feasibility Analysis		Date:
Student Names:	1. 2. 3. 4.	

Objectives : Given the materials, tools and equipment, upon completion of the activity, the student must be able to:

1. Analyze technical feasibility of the proposed project.
2. Identify the associated risks.

Background :

Feasibility Analysis

Once the need for the system and its business requirements have been defined, the approval committee may authorize the systems analyst to prepare a more detailed business case to better understand the proposed information system project. **Feasibility analysis** guides the organization in determining whether to proceed with the project. Feasibility analysis also identifies the important **risks** associated with the project that must be managed if the project is approved. As with the system request, each organization has its own process and format for the feasibility analysis, but most include techniques to assess three areas: technical feasibility, economic feasibility, and organizational feasibility. The results of evaluating these three feasibility factors are combined into a *feasibility study* deliverable that is submitted to the approval committee at the end of project initiation.

Technical Feasibility

The first technique in the feasibility analysis is to assess the *technical feasibility* of the project, the extent to which the system can be successfully designed, developed, and installed by the IT group. Technical feasibility analysis is, in essence, a *technical risk analysis* that strives to answer the question: “Can we build it?”⁷

Technical Feasibility: Can We Build It?

- Familiarity with application: Less familiarity generates more risk.
- Familiarity with technology: Less familiarity generates more risk.
- Project size: Large projects have more risk.
- Compatibility: The harder it is to integrate the system with the company’s existing technology, the higher the risk will be.

Many **risks** can endanger the successful completion of the project. First and foremost is

1. The users' and analysts' **familiarity with the application**. When analysts are unfamiliar with the business application area, they have a greater chance of misunderstanding the users or missing opportunities for improvement. The risks increase dramatically when the users themselves are less familiar with an application, such as with the development of a system to support a new business innovation (e.g., Microsoft starting up a new Internet dating service). In general, the development of new systems is riskier than extensions to an existing system, because existing systems tend to be better understood.
2. **Familiarity with the technology** is another important source of technical risk. When a system will use technology that has not been used before *within the organization*, there is a greater chance that problems and delays will occur because of the need to learn how to use the technology. Risk increases dramatically when the technology itself is new (e.g., web development using Ajax).
3. **Project size** is an important consideration, whether measured as the number of people on the development team, the length of time it will take to complete the project, or the number of distinct features in the system. Larger projects present more risk, because they are more complicated to manage and because there is a greater chance that some important system requirements will be overlooked or misunderstood. The extent to which the project is highly integrated with other systems (which is typical of large systems) can cause problems, because complexity is increased when many systems must work together.
4. Finally, project teams need to consider the **compatibility** of the new system with the technology that already exists in the organization. Systems rarely are built in a vacuum—they are built in organizations that have numerous systems already in place. New technology and applications need to be able to integrate with the existing environment for many reasons. They may rely on data from existing systems, they may produce data that feed other applications, and they may have to use the company's existing communications infrastructure. A new CRM system, for example, has little value if it does not use customer data found across the organization in existing sales systems, marketing applications, and customer service systems.

The assessment of a project's technical feasibility is not cut and dried, because in many cases, some interpretation of the underlying conditions is needed (e.g., how large does a project need to grow before it becomes less feasible?). One approach is to compare the project under consideration with prior projects undertaken by the organization. Another option is to consult with experienced IT professionals in the organization or with external IT consultants; often, they will be able to judge whether a project is feasible from a technical perspective.

Resources: ebook "System Analysis & Design 5th Edition by Dennis, Wixom, and Roth pages 23-24"

Accomplish the following task:

1. Create a technical feasibility analysis of the proposed system.
(refer to the given sample as guide)

sample:

Digital Music Download Project Executive Summary

Carly Edwards and Jason Wells created the following feasibility analysis for the Tune Source Digital Music Download Project. The System Request is attached, along with the detailed feasibility study. The highlights of the feasibility analysis are as follows:

Technical Feasibility

The Digital Music Download system is feasible technically, although there is some risk.

Tune Source's risk regarding familiarity with music download applications is moderately high.

- The Marketing Department has little experience with a subscription-based business model.
- The IT department has strong knowledge of the company's existing Web-based CD sales system, but it has not worked with music downloads or customer subscriptions.
- Numerous music download sites exist on the Internet.

Tune Source's risk regarding familiarity with the technology is moderately low.

- The IT department has knowledge of the current Web-based order entry system and the databases and Internet technology it uses.
- The IT department has no direct knowledge of the technology required to store and deliver digital music downloads; however, many of the technical issues will be the responsibility of the ISP.
- Consultants are readily available to provide help in this area.

The project size is considered medium risk.

- The project team will likely consist of 10 or fewer people.
- Business user involvement will be required.
- The project time frame is somewhat critical, since the system is needed to maintain our competitive position in the market.

The compatibility with Tune Source's existing technical infrastructure should be good.

- An Internet infrastructure is already in place at the retail stores and corporate headquarters.
- The ISP should be able to scale its services to accommodate the new Digital Music Download system.